

# The American Journal of Sports Medicine

<http://ajs.sagepub.com/>

---

## Quality of Life and Clinical Outcome After Anterior Cruciate Ligament Reconstruction Using Patellar Tendon Graft or Quadrupled Semitendinosus Graft: An 8-Year Follow-up of a Randomized Controlled Trial

Björn Barenius, Martin Nordlander, Sari Ponzer, Jan Tidermark and Karl Eriksson

*Am J Sports Med* 2010 38: 1533 originally published online June 21, 2010

DOI: 10.1177/0363546510369549

The online version of this article can be found at:

<http://ajs.sagepub.com/content/38/8/1533>

---

Published by:



<http://www.sagepublications.com>

On behalf of:

American Orthopaedic Society for Sports Medicine



Additional services and information for *The American Journal of Sports Medicine* can be found at:

Email Alerts: <http://ajs.sagepub.com/cgi/alerts>

Subscriptions: <http://ajs.sagepub.com/subscriptions>

# Corr ectio n

A correction has been published for this article. The contents of the [correction](#) have been appended to the original article in this reprint. The correction is also available online at:

<http://ajs.sagepub.com/content/39/4/NP6.full.html>

Reprints: <http://www.sagepub.com/journalsReprints.nav>

Permissions: <http://www.sagepub.com/journalsPermissions.nav>

>> [Version of Record](#) - Jul 30, 2010

[OnlineFirst Version of Record](#) - Jun 21, 2010

[What is This?](#)

Downloaded from [ajs.sagepub.com](http://ajs.sagepub.com) by guest on November 24, 2014

# Quality of Life and Clinical Outcome After Anterior Cruciate Ligament Reconstruction Using Patellar Tendon Graft or Quadrupled Semitendinosus Graft

## An 8-Year Follow-up of a Randomized Controlled Trial

Björn Barenius,<sup>\*†</sup> MD, Martin Nordlander,<sup>‡</sup> PT, Sari Ponzer,<sup>†</sup> MD, PhD,  
Jan Tidermark,<sup>†§</sup> MD, PhD, and Karl Eriksson,<sup>†</sup> MD, PhD

*From the <sup>†</sup>Section of Orthopaedics, Department of Clinical Science and Education, Södersjukhuset, Karolinska Institutet, Stockholm, Sweden, <sup>‡</sup>Section of Physiotherapy and Occupational Therapy, Emergency Department, Södersjukhuset, Stockholm, Sweden, and <sup>§</sup>Department of Orthopaedics, Capio St Görans Hospital, Stockholm, Sweden*

**Background:** Randomized controlled trials after anterior cruciate ligament reconstructions with long-term follow-up including assessment of health-related quality of life are rare.

**Purpose:** To compare clinical outcome and health-related quality of life 8 years after anterior cruciate ligament reconstruction using 2 types of graft.

**Study Design:** Randomized controlled trial; Level of evidence, 1.

**Methods:** Long-term follow-up of 164 patients with anterior cruciate ligament injury randomized to arthroscopic reconstruction with a quadrupled semitendinosus graft or a bone–patellar tendon–bone graft was undertaken. After a mean 8 years, 153 patients were available for follow-up, including instrumented laxity, 1-legged hop test, a knee-walking test, and assessment with International Knee Documentation Committee, Lysholm, Tegner, and patellofemoral pain score. Health-related quality of life was assessed with Knee Osteoarthritis Outcome Score and Short Form–36.

**Results:** Patients in both graft groups retained the same stability, knee function, and health-related quality of life. The patellofemoral pain score was similar for both groups; the bone–patellar tendon–bone group had more donor site morbidity from kneeling and knee walking. In the bone–patellar tendon–bone group, 19 patients had no kneeling problems, 23 slight problems, 31 moderate problems, and 5 unable to kneel. Corresponding figures for the semitendinosus group were 25, 32, 16, and 2 ( $P < .001$ ). Patients with early reconstructions ( $<5$  months) had a lower risk for meniscal injuries (37%) than did later reconstructed (62%,  $P = .008$ ). Health-related quality of life regarding physical functioning in Short Form–36 was better for the early-reconstructed patients than for the later reconstructed (92 vs 85;  $P = .014$ ). Patients without medial meniscal surgery had higher Knee Osteoarthritis Outcome Scores for all subscales than did patients with medial meniscal surgery, with most significant difference for sport and recreation (63 vs 75,  $P = .008$ ).

**Conclusion:** In the long term, the semitendinosus graft provided similar stability, knee function, and health-related quality of life but with less kneeling morbidity than did the bone–patellar tendon–bone graft.

**Keywords:** anterior cruciate ligament (ACL); patellar tendon graft; bone–patellar tendon–bone (BPTB); semitendinosus tendon (ST) graft; health-related quality of life (HRQoL); long-term follow-up; randomized controlled trial (RCT)

\*Address correspondence to Björn Barenius, MD, Department of Orthopaedics, Södersjukhuset, SE-11883 Stockholm, Sweden (e-mail: bjorn.barenius@sodersjukhuset.se).

The authors declared that they had no conflicts of interests in their authorship and publication of this contribution.

During the past decades, ACL reconstruction with a bone–patellar tendon–bone (BPTB) graft has been regarded as the procedure of choice,<sup>7,47</sup> but later studies have shown that BPTB reconstruction can result in graft-related morbidity with anterior knee pain and kneeling pain.<sup>14,20</sup> During the past decade, there has been a shift from BPTB as the most common graft choice in favor of a hamstring graft, that is, a semitendinosus/gracilis tendon (ST/G) graft or an

ST graft without gracilis. However, the debate about which graft provides the best long-term results is still ongoing, and since the year 2001, at least 15 comparisons between BPTB and the quadrupled ST/G graft have been published<sup>11</sup>; since the year 2005, 4 meta-analyses of BPTB versus hamstring graft have been published.<sup>3-5,14</sup> Earlier studies on the ST graft are more rare, there being only 3 comparing ST and BPTB grafts<sup>8,9,24</sup> and 1 comparing ST and ST/G grafts.<sup>13</sup> Interestingly, although this shift in clinical practice has taken place, there are only a few studies with a longer follow-up than 5 years comparing BPTB and hamstring grafts, all ST/G grafts,<sup>24,31,34,37,38</sup> and only 3 of these studies are randomized controlled trials (RCTs).<sup>16,24,38</sup>

In recent years, health-related quality of life (HRQoL) measures have been introduced more frequently in orthopaedic outcome studies. Because these instruments report the outcome from the patient's perspective, they serve as an important complement to other outcome measures, for example, traditional knee scores,<sup>30,35</sup> and contribute to a more complete picture of how the injury/disease influences all areas of life. This offers an opportunity to compare the outcome for patients with different injuries/diseases affecting the musculoskeletal system and to compare the outcome with that of patients suffering from injuries/diseases not solely affecting the musculoskeletal system.

The present study is an 8-year follow-up of an RCT comparing BPTB and ST grafts with previously published 2-year results.<sup>8</sup> The aim was to compare the clinical outcome and HRQoL 8 years after an ACL reconstruction in patients randomized to a BPTB or an ST graft. The hypothesis was that patients with an ST graft would perform as well or better over time than would patients with a BPTB graft with regard to knee function, graft-related morbidity, and HRQoL.

## PATIENTS AND METHODS

From 1995 to 1997, 164 patients were included in a 2-center RCT.<sup>8</sup> The randomization was done with numbered, independently prepared opaque envelopes. Eighty-four patients were randomized to ACL reconstruction with an ipsilateral BPTB graft (interference screw fixation on the tibial and femoral sides) and 80 patients to an ipsilateral ST graft (EndoButton on the femoral side and suture-over-post fixation on the tibia), according to the standards of the time 1995 to 1997. For further details regarding surgical technique, we refer to previously published data.<sup>8,9</sup> Inclusion criteria were single-leg ACL insufficiency with the trauma occurring at least 2 months before reconstruction, age between 15 and 45 years, no previous cruciate ligament reconstruction in the involved knee, and no concomitant posterior cruciate ligament insufficiency. Before surgery, the patients were evaluated using the Lysholm score,<sup>27</sup> the Tegner activity level score,<sup>44</sup> and

International Knee Documentation Committee (IKDC) form.<sup>17</sup> Knee surgeons well experienced in both methods performed the surgery, and the same rehabilitation protocol was used for all patients. The local ethics committee approved the study, and all patients gave their informed consent to participate.

The 2-year assessment<sup>8</sup> showed equal results in both groups regarding stability and knee function, except that patients in the ST group had better kneeling ability. The overall conclusion after 2 years was that BPTB and quadrupled ST grafts resulted in a similar outcome in the medium term. Moreover, associated meniscal injury had an adverse effect on outcome, and early reconstruction (operation performed <5 months after injury) seemed to be beneficial.

For the present study, all patients were contacted again 7 to 10 years after the initial surgery. Of the 164 initially included patients, 153 (93%) participated (BPTB group,  $n = 78$ ; ST group,  $n = 75$ ). Two subjects had moved abroad, 2 could not be reached, and 7 chose not to participate. The mean time to follow-up was 8.4 years (SD, 0.98).

Our primary analysis focused on reporting the long-term outcome in relation to randomization. As an additional analysis, we also compared the outcome for patients having reconstructions within 5 months after injury (early reconstruction,  $n = 46$ ) to the outcome of those having reconstructions later (late reconstruction,  $n = 107$ ), regardless of the graft used. Moreover, we analyzed the outcome in relation to presence of a meniscal injury (meniscal injury,  $n = 83$ ; no meniscal injury,  $n = 70$ ). Meniscal injury was defined as an injury diagnosed at the initial arthroscopy and/or at index surgery, regardless of what treatment the injury necessitated. We also analyzed the subgroups having any kind of surgical interventions due to their meniscal injuries compared with the subgroup without meniscal injury at the inclusion arthroscopy or the index surgery.

## Methods

The clinical examinations were conducted by an independent examiner (M. N.), a physical therapist not involved in the surgery or the rehabilitation but not blinded to the type of intervention. In all clinical tests, the contralateral, initially noninjured side was used as a reference.

The follow-up included an assessment using the Lysholm score,<sup>27</sup> Tegner activity level score,<sup>44</sup> IKDC score,<sup>17</sup> and the patellofemoral pain score according to Werner et al.<sup>49</sup> Moreover, a knee-walking test, as described by Kartus et al,<sup>20</sup> was used, and the area with disturbed sensitivity at the harvest site was measured. In addition to the knee-walking test, we also asked the question, "Do you experience a sensory loss?" to assess the patient's subjectively experienced sensory loss. The subjects answered on a graded scale with 3 levels: no sensory loss, disturbed sensitivity, and lost sensitivity. All instruments were self-administered by the patients.

Functional strength was assessed with the 1-legged hop test,<sup>29</sup> and laxity was evaluated with the Lachman test, with the pivot-shift test, and by measuring with a Rolimeter,<sup>1</sup>

<sup>11</sup>References 2, 6, 10, 19, 22-24, 31, 34, 37-41, 45.

using a manual maximum force. Range of motion (ROM) was reported as a deficit in extension and flexion.

In addition to these knee-specific measurements, the patients rated their knee function after the reconstruction on a visual analog scale (VAS: 0, worst possible score; 100, best possible score) by answering the following 2 questions: "How does your knee function?" (VAS 1) and "How does your knee affect your activity level?" (VAS 2). Furthermore, the patients were also asked to rate their HRQoL according to the Knee Osteoarthritis Outcome Score (KOOS)<sup>35</sup> and the Short Form-36 (SF-36).<sup>46</sup>

The KOOS is a knee-specific HRQoL instrument that was developed to assess the patients' opinions about their knees and associated problems, and it has been used previously for patients with ACL injuries.<sup>25</sup> The KOOS consists of 5 subscales: pain, other symptoms, function in daily living (ADL), function in sport and recreation, and knee-related quality of life (QoL).<sup>35</sup>

The SF-36 is an HRQoL instrument comprising 36 statements concerning physical and mental function aggregated into 8 subscales, that is, physical functioning, role limitations due to physical function, bodily pain, general health, vitality, social functioning, role limitations due to emotional problems, and mental health.<sup>43</sup>

## Statistics

The statistical software used was SPSS 17.0 for Windows. Nominal variables were tested by the  $\chi^2$  test or the Fisher exact test. Ordinal variables and nonnormality distributed interval and ratio scale variables were evaluated by the Mann-Whitney *U* test, and the Student *t* test was used for normally distributed interval and ratio scale variables in independent groups. Longitudinal statistics were done with the paired-samples *t* test for normally distributed scale variables and the Wilcoxon signed rank test for ordinal and nonnormality distributed scale variables. The tests were 2-sided. The results were considered significant at  $P < .05$ .

## RESULTS

### Baseline Data

The randomization groups were comparable regarding most baseline data, that is, weight, time to reconstruction, IKDC preoperatively, Tegner score and Lysholm score preoperatively, and VAS 1 and VAS 2 preoperatively. One difference applied to gender; there were somewhat more men in the ST group ( $P = .016$ ). The other difference pertained to age; the ST group was older than was the BPTB group ( $P = .021$ ) (Table 1).

The 2 subgroups of patients having early reconstructions and those with late reconstructions were similar regarding baseline data, except for time to reconstruction according to the grouping criteria and Tegner activity score, with a median of 9 (range, 3-10) versus 8 (range, 2-10;  $P = .001$ ). The mean waiting time to reconstruction

TABLE 1  
Demographic Data at 8 Years and Additional Procedures After the Index Procedure (Index Knee) in Relation to Randomization Groups<sup>a</sup>

|                                                                  | BPTB Group<br>(n = 78) | ST Group<br>(n = 75) | P    |
|------------------------------------------------------------------|------------------------|----------------------|------|
| Mean (SD) age at follow-up, y                                    | 33 (6.3)               | 35 (7.5)             | .021 |
| Mean (SD) weight increase, kg                                    | 4.7 (6.6)              | 5.7 (7.9)            | .41  |
| Male gender (%)                                                  | 38 (42.7)              | 51 (57.3)            | .016 |
| Total number of additional procedures after the index procedure  | 23                     | 23                   | .83  |
| Arthroscopy and shaving                                          | 11                     | 9                    |      |
| Meniscal surgery                                                 | 9                      | 5                    |      |
| Graft rupture with revision                                      | 2                      | 2                    |      |
| Biopsy                                                           |                        | 1                    |      |
| Arthroscopy and notchplasty                                      | 1                      |                      |      |
| Chondrocyte transplantation                                      |                        | 1                    |      |
| Osteotomy                                                        |                        | 1                    |      |
| Osteosynthesis of tibia fracture                                 |                        | 1                    |      |
| Extraction of tibia post (screw)                                 |                        | 3                    |      |
| Total number of additional procedures after the 2-year follow-up | 18                     | 15                   | .70  |
| Arthroscopy and shaving                                          | 9                      | 3                    |      |
| Meniscal surgery                                                 | 6                      | 4                    |      |
| Graft rupture with revision                                      | 2                      | 2                    |      |
| Biopsy                                                           |                        | 1                    |      |
| Arthroscopy and notchplasty                                      | 1                      |                      |      |
| Chondrocyte transplantation                                      |                        | 1                    |      |
| Osteotomy                                                        |                        | 1                    |      |
| Osteosynthesis of tibia fracture                                 |                        | 1                    |      |
| Extraction of tibia post (screw)                                 |                        | 2                    |      |

<sup>a</sup>BPTB, bone-patellar tendon-bone; ST, semitendinosus tendon.

was 3.6 months (SD, 1.1) for the early reconstruction group compared with 21.3 months (SD, 27.3) for those with late reconstructions ( $P < .001$ ).

The 2 subgroups of patients with and without a meniscal injury were similar regarding baseline data, except that there was a significant difference in the waiting time to reconstruction with a mean of 12 months (SD, 14) for patients without a meniscal injury compared with 20 months (SD, 30) for those with a meniscal injury ( $P = .033$ ).

The 2 subgroups of patients who underwent meniscal surgery and those who did not were similar regarding baseline data, except that there was a significant difference in the waiting time to reconstruction with a mean of 13 months (SD, 18) for patients without medial meniscal surgery compared with 22 months (SD, 32) for those with medial meniscal surgery ( $P = .019$ ).

The 11 subjects who did not attend the follow-up did not differ from the subjects who attended with regard to baseline and surgical data or outcome from earlier follow-ups.

### Surgical Outcome

The number of surgical interventions performed during follow-up did not differ between the randomization groups

TABLE 2

Range of Motion and Laxity After 8 Years in Relation to Randomization Groups<sup>a</sup>

|                                                           | BPTB Group<br>(n = 78) | ST Group<br>(n = 75) | P   |
|-----------------------------------------------------------|------------------------|----------------------|-----|
| Extension deficit, N (%)                                  |                        |                      | .24 |
| <3°                                                       | 59 (76)                | 63 (84)              |     |
| 3°-5°                                                     | 19 (24)                | 10 (13)              |     |
| 6°-10°                                                    | 0 (0)                  | 3 (3)                |     |
| Flexion deficit, N (%)                                    |                        |                      | .13 |
| 0°-5°                                                     | 72 (97)                | 69 (92)              |     |
| 6°-25°                                                    | 6 (3)                  | 6 (8)                |     |
| Lachman, N (%)                                            |                        |                      | .61 |
| 0 (0-2)                                                   | 48 (61)                | 50 (67)              |     |
| +1 (3-5)                                                  | 28 (36)                | 21 (28)              |     |
| +2 (6-10)                                                 | 0 (0)                  | 3 (4)                |     |
| +3 (>10)                                                  | 2 (3)                  | 1 (1)                |     |
| Pivot shift, N (%)                                        |                        |                      | .35 |
| 0                                                         | 62 (80)                | 54 (72)              |     |
| +1                                                        | 12 (15)                | 19 (25.3)            |     |
| +2                                                        | 1 (1)                  | 1 (1.3)              |     |
| +3                                                        | 0 (0)                  | 1 (1.3)              |     |
| Not possible to perform                                   | 3 (4)                  | 0 (0)                |     |
| Mean (SD) Rolimeter<br>maximum side-to-side<br>difference | 2.3 (1.9)              | 2.3 (1.9)            | .92 |

<sup>a</sup>BPTB, bone–patellar tendon–bone; ST, semitendinosus tendon.

(Table 1). During the follow-up period, 46 patients (28%) had had an additional procedure in the index knee (23 patients in each group), and 33 of them had surgery after the 2-year follow-up (15 ST and 18 BPTB). Four patients had been revised because of a graft rupture (2 BPTB and 2 ST). Ten patients reported a diagnosed rupture of the contralateral ACL (3 ST and 7 BPTB), 5 of which had been reconstructed.

### Outcome for the Randomization Groups at 8 Years

There were no differences in ROM or stability according to the laxity tests on comparing the randomization groups after 8 years (Table 2). The patients also reported comparable knee function according to the IKDC, the Tegner activity level score, the Lysholm score, and the VAS questions (Tables 3 and 4).

The 1-legged hop test could not be performed by 13 subjects, 5 in the BPTB group and 8 in the ST group. The reasons for not participating were problems with the contralateral knee in 3 patients, pregnancy in 2, knee pain in 4, and unknown reasons in 3. The length of the jump with the reconstructed leg compared with that of the contralateral leg, as analyzed by intention to treat, was a mean of 96.4% (SD, 9.2%) for the ST group and 96.7% (SD, 11.0%) for the BPTB group ( $P = .387$ ).

The mean total patellofemoral score according to Werner et al was 40 in both groups (0, worst possible; 55, best possible).<sup>8,49</sup> However, the BPTB group reported significantly more problems for the subscore kneeling ( $P < .001$ ). The results for the BPTB group were 19 patients

TABLE 3

Outcome According to IKDC and the Tegner Activity Level Scores Preoperatively, at 2 Years, and After 8 Years in Relation to Randomization Groups<sup>a</sup>

|                               | BPTB Group | ST Group  | P   |
|-------------------------------|------------|-----------|-----|
| IKDC, n (%)                   |            |           |     |
| Preoperatively                |            |           | .60 |
| A                             | 0 (0)      | 0 (0)     |     |
| B                             | 0 (0)      | 0 (0)     |     |
| C                             | 15 (19)    | 17 (23)   |     |
| D                             | 63 (81)    | 58 (77)   |     |
| At 2 years                    |            |           | .77 |
| A                             | 3 (4.1)    | 4 (5.7)   |     |
| B                             | 32 (43.2)  | 30 (42.9) |     |
| C                             | 24 (32.4)  | 23 (32.9) |     |
| D                             | 15 (20.3)  | 13 (18.6) |     |
| At 8 years                    |            |           | .83 |
| A                             | 7 (9)      | 11 (15)   |     |
| B                             | 45 (58)    | 36 (48)   |     |
| C                             | 16 (20)    | 21 (28)   |     |
| D                             | 10 (13)    | 7 (9)     |     |
| Difference in IKDC, P         |            |           |     |
| Preoperative vs 8 years       | .001       | <.001     | NA  |
| 2 years vs 8 years            | .023       | .003      | NA  |
| Tegner score, median (range)  |            |           |     |
| Preoperative                  | 2 (1-10)   | 2 (0-6)   | .66 |
| At 2 years                    | 6 (1-10)   | 5 (1-10)  | .32 |
| At 8 years                    | 5 (0-10)   | 4 (1-9)   | .41 |
| Difference in Tegner score, P |            |           |     |
| Preoperative vs 8 years       | <.001      | <.001     | NA  |
| 2 years vs 8 years            | <.001      | .002      | NA  |

<sup>a</sup>BPTB, bone–patellar tendon–bone; IKDC, International Knee Documentation Committee; NA, not applicable; ST, semitendinosus tendon. Subjects who participated in all follow-ups: preoperatively, BPTB, n = 78; ST, n = 75; at 2 years, BPTB, n = 74; ST, n = 70; at 8 years, BPTB, n = 78; ST, n = 75.

with no problem with kneeling, 23 with slight problems, 31 with moderate problems, and 5 unable to kneel. The corresponding figures for the ST group were 25, 32, 16, and 2. The ST group also had a significantly better result in the knee-walking test according to Kartus et al ( $P < .001$ ).<sup>20</sup> In the BPTB group, there were 24 patients with no problems, 22 with slight problems, 22 with moderate problems, and 10 with severe problems. The corresponding figures for the ST groups were 46, 17, 6, and 6.

In addition, the patients in the BPTB group reported more pronounced loss of skin sensation than did the ST group ( $P = .0019$ ). In the BPTB group, there were 21 patients without sensitivity loss, 15 with disturbed sensitivity, and 42 with lost sensitivity from the ACL reconstruction. The corresponding figures for the ST groups were 27, 24, and 24.

There were no differences between the randomization groups in reported HRQoL according to KOOS (Figure 1) and SF-36 (Figure 2).

At the final follow-up, 144 (94%) of the patients stated that they did not regret the ACL surgery, whereas 3 (2%) did regret the surgery, and 6 (4%) were unsure. Three



TABLE 4  
Outcome According to the Lysholm Score, VAS 1, and VAS 2 Preoperatively, at 2 Years, and After 8 Years in Relation to Randomization Groups<sup>a</sup>

|                                | BPTB Group             | ST Group               | P    |
|--------------------------------|------------------------|------------------------|------|
| Lysholm score, median (range)  |                        |                        |      |
| Preoperatively                 | 60 (2-90)              | 61 (2-87)              | .45  |
| At 2 years                     | 85 (46-100)            | 87 (45-100)            | .83  |
| At 8 years                     | 88 (43-100)            | 89 (40-100)            | .32  |
| Difference in Lysholm score, P |                        |                        |      |
| Preoperative vs 8 years        | <.001                  | <.001                  | NA   |
| 2 years vs 8 years             | .60                    | .18                    | NA   |
| VAS 1, median (range)          |                        |                        |      |
| Preoperatively                 | 35 (0-93) <sup>b</sup> | 27 (0-90) <sup>c</sup> | .38  |
| At 2 years                     | 81 (0-100)             | 81 (0-100)             | .60  |
| At 8 years                     | 74 (0-100)             | 73 (0-100)             | .58  |
| Difference in VAS 1, P         |                        |                        |      |
| Preoperative vs 8 years        | .001                   | .001                   | NA   |
| 2 years vs 8 years             | .97                    | .51                    | NA   |
| VAS 2, median (range)          |                        |                        |      |
| Preoperatively                 | 20 (0-82) <sup>d</sup> | 20 (0-99) <sup>c</sup> | .83  |
| At 2 years                     | 82 (0-100)             | 80 (0-100)             | .52  |
| At 8 years                     | 72 (0-100)             | 72 (0-100)             | 1.00 |
| Difference in VAS 2, P         |                        |                        |      |
| Preoperative vs 8 years        | <.001                  | <.001                  | NA   |
| 2 years vs 8 years             | .88                    | .27                    | NA   |

<sup>a</sup>BPTB, bone–patellar tendon–bone; NA, not applicable; ST, semitendinosus tendon; VAS, visual analog scale. VAS 1, How does your knee function? (100, normal; 0, worst possible function); VAS 2, How does your knee affect your level of activity? (100, normal; 0, worst possible). Subjects who participated in all follow-ups: BPTB, n = 78; ST, n = 75.

<sup>b</sup>38 missing values.

<sup>c</sup>27 missing values.

<sup>d</sup>37 missing values.

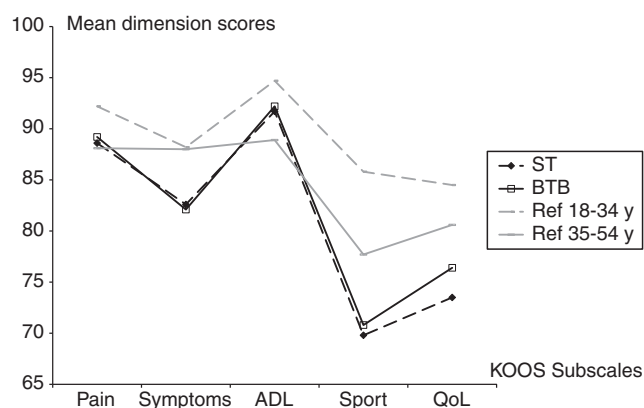
patients in the BPTB group regretted the surgery, and 3 patients in both graft groups were unsure ( $P = .314$ ).

### Long-term Compared With Medium-term Outcome in the Randomization Groups

The IKDC values (Table 3) and the Lysholm score (Table 4) continued to improve from the 2-year follow-up, whereas the overall median Tegner activity score (Table 3) and VAS ratings (Table 4) were lower at the current follow-up compared with the values at 2 years but still better compared with the preoperative scores. The results for the Lachman and pivot-shift tests did not change over time (see online Appendix for this article at <http://ajs.sagepub.com/supplemental/>).

### Outcome Comparing Early Versus Late Reconstruction

The outcome at 8 years showed that patients who had an early reconstruction (<5 months) had fewer meniscal injuries compared with those with a late reconstruction (>5



**Figure 1.** The mean KOOS at the 8-year follow-up for patients in the BPTB group (n = 78) compared with the ST group (n = 75) and, as a reference, 2 age-matched healthy-knee reference populations.<sup>30</sup> 0, extreme symptoms; 100, no symptoms. P values for differences between the BPTB and ST groups on the subscales: pain ( $P = .76$ ), symptoms ( $P = .83$ ), function in daily living (ADL,  $P = .79$ ), function in sport and recreation ( $P = .88$ ), and knee-related quality of life (QoL,  $P = .35$ ). BPTB, bone–patellar tendon–bone; KOOS, Knee Osteoarthritis Outcome Score; ST, semitendinosus.

months). Sixty-six of 107 (62%) in the late reconstruction group had meniscal injuries, compared with 17 of 46 (37%) in the early reconstruction group ( $P = .008$ ).

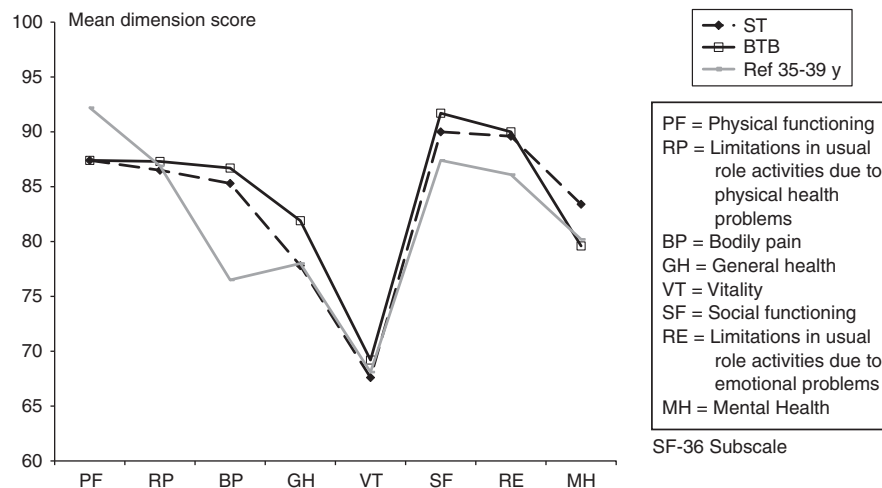
Moreover, the early reconstruction patients had a higher Tegner activity level at 8 years compared with those with later reconstructions: median of 6 (range, 1-10) and 3 (range, 1-9), respectively ( $P < .001$ ).

The KOOS ratings for the early reconstruction group showed higher figures than for late reconstruction in all subscales, but the difference was statistically significant only for the subscale ADL and with a trend for QoL (Figure 3). Furthermore, the early reconstruction patients also scored better on the SF-36 subscales physical functioning, bodily pain, and social functioning compared with those with later reconstruction (Figure 4).

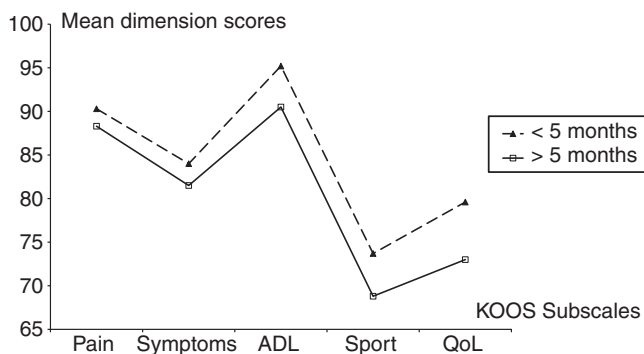
The results for ROM, laxity, pivot shift, 1-legged hop, IKDC score, Lysholm score, patellofemoral score, and patients' satisfaction according to the VAS were similar for the 2 groups (see online Appendix for this article at <http://ajs.sagepub.com/supplemental/>).

### Outcome for Patients With or Without a Meniscal Injury at Inclusion

Patients with a meniscal injury at inclusion, compared with those without, displayed increased knee laxity at the 8-year follow-up; that is, more often they had a positive and more pronounced pivot-shift result compared with those without. Among patients with a meniscal injury, 27% had a grade 1 pivot shift and 4% grade 2 or more compared with only 13% with a grade 1 pivot shift among patients without a meniscal injury ( $P = .008$ ). The difference in favor of patients without a meniscal injury at



**Figure 2.** The mean SF-36 scores at the 8-year follow-up for patients in the BPTB group ( $n = 78$ ) compared with the ST group ( $n = 75$ ). As a reference, an age-matched healthy Swedish population is included.<sup>42</sup> 0, worst possible score; 100, best possible score.  $P$  values for differences between the BPTB and ST groups on the subscales: PF ( $P = .99$ ), RP ( $P = .85$ ), BP ( $P = .71$ ), GH ( $P = .16$ ), VT ( $P = .62$ ), SF ( $P = .56$ ), RE ( $P = .93$ ), and MH ( $P = .15$ ). BPTB, bone-patellar tendon-bone; SF-36, Short Form-36; ST, semitendinosus.



**Figure 3.** The mean Knee Osteoarthritis Outcome Score (KOOS) at the 8-year follow-up for patients who had an early ( $n = 46$ ) or a late ( $n = 107$ ) reconstruction. 0, extreme symptoms; 100, no symptoms.  $P$  values for differences between patients who had an early or late reconstruction on the subscales: pain ( $P = .37$ ), symptoms ( $P = .33$ ), function in daily living (ADL,  $P = .029$ ), function in sport and recreation ( $P = .23$ ), and knee-related quality of life (QoL,  $P = .059$ ).

inclusion was also apparent in the objective IKDC ratings. The objective IKDC results for patients without a meniscal injury were 35 A, 31 B, 4 C, and no D. The corresponding ratings for patients with a meniscal injury were 22 A, 53 B, 4 C, and 4 D ( $P = .003$ ). There was a trend toward more extension deficit in patients in the meniscal injury group, although not significant ( $P = .083$ ). Otherwise, there were no significant differences between the groups regarding ROM, laxity, 1-legged hop, Tegner activity score, Lysholm score, the total IKDC score, or patients' satisfaction (VAS).

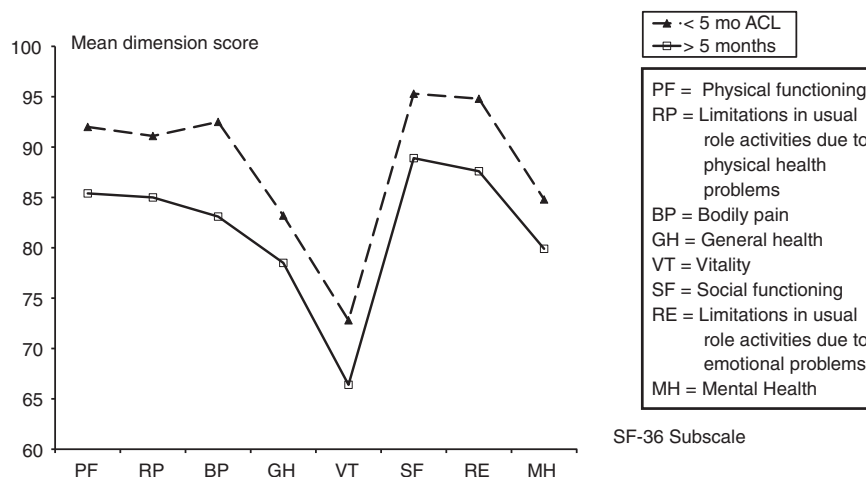
The KOOS ratings for patients without a meniscal injury showed higher figures than for patients with

a meniscal injury in all subscales, but the difference was not statistically significant. The SF-36 scores for patients without a meniscal injury, compared with those for patients with a meniscal injury for all subscales, revealed no significant differences.

### Outcome for Patients Having Surgical Intervention Because of Meniscal Injury

In patients who underwent medial meniscal surgery, the pivot shift was more common ( $P = .016$ ) but not for the other subgroups. The KOOS showed significant differences in favor of patients without medial meniscal surgery ( $n = 92$ ) compared with those with medial meniscal surgery ( $n = 60$ ) in the subscales symptoms (85 vs 79,  $P = .027$ ), ADL (94 vs 88,  $P = .005$ ), function in sport and recreation (75 vs 63,  $P = .008$ ), and QoL (78 vs 71,  $P = .035$ ). The SF-36 showed a significantly better value for patients without medial meniscal surgery only in the subscale physical functioning, with a score of 90 compared with 84 for patients with medial meniscal surgery ( $P = .036$ ). The group with medial meniscal surgery was also worse compared with the group without medial meniscal surgery on the Lysholm score, with a median of 84 (range, 40-100) versus 89 (range, 43-100;  $P = .021$ ), and the difference between reconstructed and uninjured in Rolimeter laxity, 2.68 mm (SD, 2.2) versus 2.06 mm (SD, 1.6;  $P = .044$ ). The reported activity level by Tegner at the current follow-up showed a trend with higher figures for the group without medial meniscal surgery, with a median of 5 (range, 1-10) versus a median of 4 (range, 0-9) for the group with medial meniscal surgery ( $P = .054$ ).

Other subgroup analyses regarding surgical intervention on the lateral meniscus or combinations of surgical interventions on lateral and/or medial menisci did not significantly differ between the groups.



**Figure 4.** The mean Short Form-36 (SF-36) scores at the 8-year follow-up for patients who had an early ( $n = 46$ ) or a late ( $n = 107$ ) reconstruction. 0, worst possible score; 100, best possible score.  $P$  values for differences between patients who had an early or late reconstruction on the subscales: PF ( $P = .014$ ), RP ( $P = .21$ ), BP ( $P = .013$ ), GH ( $P = .14$ ), VT ( $P = .065$ ), SF ( $P = .037$ ), RE ( $P = .12$ ), and MH ( $P = .075$ ).

## DISCUSSION

This study reports several major findings concerning the long-term results after ACL reconstruction. First, the results of the randomization showed that patients with the ST graft retained the same stability and knee function according to the Lysholm and IKDC functional scores as did the BPTB graft over an 8-year period with less graft-specific morbidity, such as problems with kneeling and discomfort from loss of sensitivity. Second, the long-term follow-up revealed that although there was a slight deterioration in knee function according to some functional scores between the 2- and 8-year follow-ups, the vast majority of the patients, 94%, did not regret the surgery. The additional analysis comparing patients with early reconstructions (<5 months) with those with later reconstructions indicated that early reconstruction may result in a lower risk for meniscal injuries, better knee function, and better HRQoL. Finally, the study confirms that an ACL injury in combination with a meniscal injury that requires surgical intervention is a negative prognostic factor regarding stability, knee function, and HRQoL.

The finding that patients with an ST graft retained the same laxity and function as did patients with a BPTB graft over the mean 8-year follow-up is in line with previously published RCTs with a minimum 5-year follow-up.<sup>16,24,38</sup> Moreover, the frequency of graft ruptures was comparable.<sup>16,24,38</sup> However, the activity level according to Tegner at the final follow-up was lower in our population compared with the level reported by Liden et al<sup>24</sup> and Harilainen et al.<sup>16</sup> This difference may be partly explained by the higher mean age in our study. Contrary to Liden et al,<sup>24</sup> we found a higher degree of donor site morbidity among patients with the BPTB graft. This difference may be explained by the 2-incision technique for harvesting the patellar tendon graft used in the study by Liden et al<sup>24</sup> compared with the single-incision technique used for the majority of our patients because the single-incision

technique potentially increases the risk for infrapatellar nerve damage. Harvesting the patellar tendon graft with a 2-incision technique may improve the kneeling morbidity associated with the BPTB graft. Despite the differences in donor site morbidity, there were no differences in HRQoL between the graft groups. A comparison with previous RCTs regarding HRQoL is not possible because none of the studies reported HRQoL.<sup>16,24,38</sup>

In a meta-analysis, Biau et al<sup>3</sup> found that the BPTB graft gives better stability as measured with the pivot shift but no differences between BPTB and ST/G regarding Lachman, which is in contrast to our finding of similar stability in both graft groups. This difference could be explained by a size effect of the graft. Although Biau et al compared ST/G grafts with 4 strands to BPTB, there is a size difference between a quadrupled ST/G and a BPTB. Our quadrupled ST grafts were a mean 9 mm in size, which might explain why they are comparable with the BPTB graft in stiffness.<sup>9</sup>

Möller et al<sup>28</sup> examined a smaller group of BPTB-reconstructed subjects in a long-term follow-up, and their results according to KOOS were similar to ours. However, they did not find any significant differences in any of the SF-36 subscales compared with a Swedish reference population. Their conclusion was that KOOS in general, and specifically the subscales of sport and recreation and QoL, seems to be more sensitive than are the SF-36 and Lysholm scores.<sup>28</sup> The individual patient's coping mechanisms may be an important factor in the patient's perception of his or her HRQoL. Langford et al<sup>21</sup> and Webster et al<sup>48</sup> have recently demonstrated the importance of the psychological response for the patients' ability to return to sport after an ACL reconstruction.

On comparing medium term to long term, we found an increase in the IKDC and Lysholm scores in spite of a marked decrease in the Tegner activity level. One explanation for this finding may be an adjustment to and/or



acceptance of a lower activity level and/or a psychological reorientation with better coping with an injured knee, which make the IKDC and Lysholm scores less sensitive for assessing changes over time.<sup>28,32</sup> Another explanation may be that the Tegner activity scores decrease for reasons unrelated to the potential function of the knee, that is, by factors unrelated to knee function such as a different social setting in midlife with less time for competitive sports and more focus on family and career.

After 8 years, the subjects, regardless of graft type, had a decreased level of activity, VAS, and quality of life, especially regarding physical function, that is, sports. Despite these findings, 94% still do not regret their surgeries. We think that the subjects appreciate that their reconstructions have given them some well-functioning years at a higher activity level than they might have had with non-operative treatment. It also seems that the VAS questions regarding knee function are more sensitive to changes over time than is the IKDC or Lysholm score because the decrease in VAS values correlates better with the decrease in the activity level and quality of life assessment.

The subgroup of patients who had their reconstructions early, within 5 months, had a significantly higher Tegner activity level score after 8 years than did those who had their reconstructions later, and they had better results regarding KOOS and SF-36. This supports the conclusion that subjects with early reconstructions retain a better knee function. However, because this is a subgroup analysis, the results should be interpreted with some caution. Isberg<sup>18</sup> found that knee kinematics measured by radiostereometric analysis gradually deteriorate during the first year after injury. Increased laxity of other stabilizing structures of the knee may therefore predispose a worse result after late ACL reconstruction with less potential to restore normal knee kinematics. Repeated pivoting episodes with bone bruising, chondral injury, and meniscal injury could also cause long-term morbidity, less resistance to higher loads, and possibly osteoarthritis.<sup>11,26</sup>

The effect of meniscal injury on the incidence of osteoarthritis is well known.<sup>12,36</sup> The patients reconstructed early also had fewer meniscal injuries, which probably influences the long-term outcome.<sup>15,33,36</sup> The meniscal injury at inclusion correlated with both a higher degree and a higher frequency of pivot shift at follow-up. One hypothesis is that the loss of the meniscus' stabilizing function makes the knee more susceptible to small traumas with a progressing laxity as a result. The group of patients with medial meniscal injuries requiring surgical intervention had a marked decrease in the KOOS subscales symptoms, ADL, sport and recreation, and QoL, a finding that is in conformity with the results of Lohmander et al.<sup>25</sup> The benefit of early reconstruction in our population can to some extent probably be explained by the morbidity of a meniscal injury.

The strength of this study is the RCT design, the long follow-up time, the high follow-up rate of 93%, the independent observer assessment, and the self-administered HRQoL and functional rating.

The weakness of the study is the lack of radiographic evaluations that might have correlated with the patient-reported outcome scores.

## CONCLUSION

The conclusion drawn from this study was that an ACL reconstruction with an ST graft stood the test of time and is a better choice than is the BPTB graft. During the 8-year follow-up, the ST graft provided stability, knee function, and HRQoL similar to that of the BPTB graft but with less kneeling morbidity. However, there was a slight deterioration of knee function with time, and the consequences of the ACL injury were still present for many of the subjects after 8 years, although the vast majority did not regret the reconstructive procedure. Additional findings of our study were that a meniscal injury influenced knee function and HRQoL negatively and that early reconstructive surgery appeared to be beneficial.

## REFERENCES

1. Balasch H, Schiller M, Friebe H, Hoffmann F. Evaluation of anterior knee joint instability with the Rolimeter: a test in comparison with manual assessment and measuring with the KT-1000 arthrometer. *Knee Surg Sports Traumatol Arthrosc.* 1999;7(4):204-208.
2. Beynon BD, Johnson RJ, Fleming BC, et al. Anterior cruciate ligament replacement: comparison of bone-patellar tendon-bone grafts with two-strand hamstring grafts: a prospective, randomized study. *J Bone Joint Surg Am.* 2002;84(9):1503-1513.
3. Biau DJ, Katsahian S, Kartus J, et al. Patellar tendon versus hamstring tendon autografts for reconstructing the anterior cruciate ligament: a meta-analysis based on individual patient data. *Am J Sports Med.* 2009;37:2470-2478.
4. Biau DJ, Tournoux C, Katsahian S, Schranz P, Nizard R. ACL reconstruction: a meta-analysis of functional scores. *Clin Orthop Relat Res.* 2007;458:180-187.
5. Biau DJ, Tournoux C, Katsahian S, Schranz PJ, Nizard RS. Bone-patellar tendon-bone autografts versus hamstring autografts for reconstruction of anterior cruciate ligament: meta-analysis. *BMJ.* 2006;332(7548):995-1001.
6. Ejerhed L, Kartus J, Sernert N, Kohler K, Karlsson J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction? A prospective randomized study with a two-year follow-up. *Am J Sports Med.* 2003;31(1):19-25.
7. Engebretsen L, Benum P, Fasting O, Molster A, Strand T. A prospective, randomized study of three surgical techniques for treatment of acute ruptures of the anterior cruciate ligament. *Am J Sports Med.* 1990;18(6):585-590.
8. Eriksson K, Anderberg P, Hamberg P, et al. A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Br.* 2001;83(3):348-354.
9. Eriksson K, Anderberg P, Hamberg P, Olerud P, Wredmark T. There are differences in early morbidity after ACL reconstruction when comparing patellar tendon and semitendinosus tendon graft: a prospective randomized study of 107 patients. *Scand J Med Sci Sports.* 2001;11(3):170-177.
10. Feller JA, Webster KE. A randomized comparison of patellar tendon and hamstring tendon anterior cruciate ligament reconstruction. *Am J Sports Med.* 2003;31(4):564-573.
11. Frobell RB, Roos HP, Roos EM, et al. The acutely ACL injured knee assessed by MRI: are large volume traumatic bone marrow lesions a sign of severe compression injury? *Osteoarthritis Cartilage.* 2008;16(7):829-836.
12. Gillquist J, Messner K. Anterior cruciate ligament reconstruction and the long-term incidence of gonarthrosis. *Sports Med.* 1999;27(3):143-156.
13. Gobbi A, Domzalski M, Pascual J, Zanazzo M. Hamstring anterior cruciate ligament reconstruction: is it necessary to sacrifice the gracilis? *Arthroscopy.* 2005;21(3):275-280.

14. Goldblatt JP, Fitzsimmons SE, Balk E, Richmond JC. Reconstruction of the anterior cruciate ligament: meta-analysis of patellar tendon versus hamstring tendon autograft. *Arthroscopy*. 2005;21(7):791-803.
15. Granan LP, Bahr R, Lie SA, Engebretsen L. Timing of anterior cruciate ligament reconstructive surgery and risk of cartilage lesions and meniscal tears: a cohort study based on the Norwegian National Knee Ligament Registry. *Am J Sports Med*. 2009;37(5):955-961.
16. Harilainen A, Linko E, Sandelin J. Randomized prospective study of ACL reconstruction with interference screw fixation in patellar tendon autografts versus femoral metal plate suspension and tibial post fixation in hamstring tendon autografts: 5-year clinical and radiological follow-up results. *Knee Surg Sports Traumatol Arthrosc*. 2006;14(6):517-528.
17. Hefti F, Muller W, Jakob RP, Staubli HU. Evaluation of knee ligament injuries with the IKDC form. *Knee Surg Sports Traumatol Arthrosc*. 1993;1(3-4):226-234.
18. Isberg J. *Kinematics and Laxity in the Knee, Before and After Anterior Cruciate Ligament Reconstruction [thesis]*. Gothenburg, Sweden: Department of Orthopaedics, Institute of Clinical Sciences, Sahlgrenska Academy at University of Gothenburg; 2008.
19. Jansson KA, Linko E, Sandelin J, Harilainen A. A prospective randomized study of patellar versus hamstring tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med*. 2003;31(1):12-18.
20. Kartus J, Ejerhed L, Sernert N, Brandsson S, Karlsson J. Comparison of traditional and subcutaneous patellar tendon harvest: a prospective study of donor site-related problems after anterior cruciate ligament reconstruction using different graft harvesting techniques. *Am J Sports Med*. 2000;28(3):328-335.
21. Langford J, Webster KE, Feller JA. A prospective longitudinal study to assess psychological changes following anterior cruciate ligament reconstruction surgery. *Br J Sports Med*. 2009;43(5):377-378.
22. Laxdal G, Kartus J, Hansson L, Heidvall M, Ejerhed L, Karlsson J. A prospective randomized comparison of bone-patellar tendon-bone and hamstring grafts for anterior cruciate ligament reconstruction. *Arthroscopy*. 2005;21(1):34-42.
23. Laxdal G, Sernert N, Ejerhed L, Karlsson J, Kartus JT. A prospective comparison of bone-patellar tendon-bone and hamstring tendon grafts for anterior cruciate ligament reconstruction in male patients. *Knee Surg Sports Traumatol Arthrosc*. 2007;15(2):115-125.
24. Liden M, Ejerhed L, Sernert N, Laxdal G, Kartus J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction: a prospective, randomized study with a 7-year follow-up. *Am J Sports Med*. 2007;35(5):740-748.
25. Lohmander LS, Englund PM, Dahl LL, Roos EM. The long-term consequence of anterior cruciate ligament and meniscus injuries: osteoarthritis. *Am J Sports Med*. 2007;35(10):1756-1769.
26. Lohmander LS, Osterberg A, Englund M, Roos H. High prevalence of knee osteoarthritis, pain, and functional limitations in female soccer players twelve years after anterior cruciate ligament injury. *Arthritis Rheum*. 2004;50(10):3145-3152.
27. Lysholm J, Gillquist J. Evaluation of knee ligament surgery results with special emphasis on use of a scoring scale. *Am J Sports Med*. 1982;10(3):150-154.
28. Möller E, Weidenhielm L, Werner S. Outcome and knee-related quality of life after anterior cruciate ligament reconstruction: a long-term follow-up. *Knee Surg Sports Traumatol Arthrosc*. 2009;17(7):786-794.
29. Noyes FR, Barber SD, Mangine RE. Abnormal lower limb symmetry determined by function hop tests after anterior cruciate ligament rupture. *Am J Sports Med*. 1991;19:513-518.
30. Paradowski PT, Bergman S, Sundén-Lundius A, Lohmander LS, Roos EM. Knee complaints vary with age and gender in the adult population: population-based reference data for the Knee Injury and Osteoarthritis Outcome Score (KOOS). *BMC Musculoskelet Disord*. 2006;7:38.
31. Pinczewski LA, Deehan DJ, Salmon LJ, Russell VJ, Clingeleffer A. A five-year comparison of patellar tendon versus four-strand hamstring tendon autograft for arthroscopic reconstruction of the anterior cruciate ligament. *Am J Sports Med*. 2002;30(4):523-536.
32. Risberg MA, Holm I, Steen H, Beynon BD. Sensitivity to changes over time for the IKDC form, the Lysholm score, and the Cincinnati knee score: a prospective study of 120 ACL reconstructed patients with a 2-year follow-up. *Knee Surg Sports Traumatol Arthrosc*. 1999;7(3):152-159.
33. Rockborn P, Messner K. Long-term results of meniscus repair and meniscectomy: a 13-year functional and radiographic follow-up study. *Knee Surg Sports Traumatol Arthrosc*. 2000;8(1):2-10.
34. Roe J, Pinczewski LA, Russell VJ, Salmon LJ, Kawamata T, Chew M. A 7-year follow-up of patellar tendon and hamstring tendon grafts for arthroscopic anterior cruciate ligament reconstruction: differences and similarities. *Am J Sports Med*. 2005;33(9):1337-1345.
35. Roos EM, Roos HP, Lohmander LS, Ekdahl C, Beynon BD. Knee Injury and Osteoarthritis Outcome Score (KOOS): development of a self-administered outcome measure. *J Orthop Sports Phys Ther*. 1998;28(2):88-96.
36. Roos H, Lauren M, Adalberth T, Roos EM, Jonsson K, Lohmander LS. Knee osteoarthritis after meniscectomy: prevalence of radiographic changes after twenty-one years, compared with matched controls. *Arthritis Rheum*. 1998;41(4):687-693.
37. Ruiz AL, Kelly M, Nutton RW. Arthroscopic ACL reconstruction: a 5-9 year follow-up. *Knee*. 2002;9(3):197-200.
38. Sajovic M, Vengust V, Komadina R, Tavcar R, Skaza K. A prospective, randomized comparison of semitendinosus and gracilis tendon versus patellar tendon autografts for anterior cruciate ligament reconstruction: five-year follow-up. *Am J Sports Med*. 2006;34(12):1933-1940.
39. Salmon L, Russell V, Musgrove T, Pinczewski L, Refshauge K. Incidence and risk factors for graft rupture and contralateral rupture after anterior cruciate ligament reconstruction. *Arthroscopy*. 2005;21(8):948-957.
40. Salmon LJ, Russell VJ, Refshauge K, et al. Long-term outcome of endoscopic anterior cruciate ligament reconstruction with patellar tendon autograft: minimum 13-year review. *Am J Sports Med*. 2006;34(5):721-732.
41. Shaieb MD, Kan DM, Chang SK, Marumoto JM, Richardson AB. A prospective randomized comparison of patellar tendon versus semitendinosus and gracilis tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med*. 2002;30(2):214-220.
42. Sullivan M, Karlsson J, Ware JE Jr. *SF-36 Hälsoenkät: Svensk Manual och Tolkningssguide [Swedish Manual and Interpretation Guide]*. Gothenburg, Sweden: Sahlgrenska University Hospital; 1994.
43. Sullivan M, Karlsson J, Ware JE Jr. The Swedish SF-36 Health Survey, I: evaluation of data quality, scaling assumptions, reliability and construct validity across general populations in Sweden. *Soc Sci Med*. 1995;41(10):1349-1358.
44. Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop Relat Res*. 1985;198:43-49.
45. Wagner M, Kaab MJ, Schallcock J, Haas NP, Weiler A. Hamstring tendon versus patellar tendon anterior cruciate ligament reconstruction using biodegradable interference fit fixation: a prospective matched-group analysis. *Am J Sports Med*. 2005;33(9):1327-1336.
46. Ware JE Jr. SF-36 health survey update. *Spine*. 2000;25(24):3130-3139.
47. Webb JM, Corry IS, Clingeleffer AJ, Pinczewski LA. Endoscopic reconstruction for isolated anterior cruciate ligament rupture. *J Bone Joint Surg Br*. 1998;80(2):288-294.
48. Webster KE, Feller JA, Lambros C. Development and preliminary validation of a scale to measure the psychological impact of returning to sport following anterior cruciate ligament reconstruction surgery. *Phys Ther Sport*. 2008;9(1):9-15.
49. Werner S, Arvidsson H, Arvidsson I, Eriksson E. Electrical stimulation of vastus medialis and stretching of lateral thigh muscles in patients with patello-femoral symptoms. *Knee Surg Sports Traumatol Arthrosc*. 1993;1(2):85-92.

# Erratum

Barenius B, Nordlander M, Ponzer S, Tidermark J, Eriksson K. Quality of life and clinical outcome after anterior cruciate ligament reconstruction using patellar tendon graft or quadrupled semitendinosus graft: an 8-year follow-up of a randomized controlled trial. *Am J Sports Med.* 2010;38(8):1533-1541. (Original DOI: 10.1177/0363546510369549)

In the above article, There is an error in Table 1 (page 1535). In the 3rd row of data, which shows gender distribution in the graft groups, the percentage for male distribution in the whole study population was depicted instead of the percentage of male distribution in the specified graft groups. The data for number of male patients in the groups are correct, and the error does not influence any of the other results.

The corrected row should read:

|                 | BPTB Group<br>(n = 78) | ST Group<br>(n = 75) | <i>P</i> |
|-----------------|------------------------|----------------------|----------|
| Male gender (%) | 38 (48.7)              | 51 (68)              | .016     |